

ORIGINAL ARTICLE

Bleeding Risk With Apixaban Versus Rivaroxaban: A Reference Trial Emulation Predicting the Results of COBRRA-VTE and COBRRA-AF Using US Health Care Claims

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BACKGROUND: Direct oral anticoagulants such as apixaban and rivaroxaban, have transformed the management of thromboembolic disorders. Randomized controlled trials were initiated in 2017 (COBRRA-VTE [Comparison of Bleeding Risk Between Rivaroxaban and Apixaban for the Treatment of Acute Venous Thromboembolism]) and 2021 (COBRRA-AF [Comparison of Bleeding Risk Between Rivaroxaban and Apixaban in Patients With Atrial Fibrillation]) to directly compare apixaban and rivaroxaban on the risk of bleeding in patients with venous thromboembolism (VTE) and atrial fibrillation (AF). We aimed to emulate the design and measurements of these ongoing trials and use US health care insurance claims data to predict their findings.

METHODS: In cohort studies designed to emulate COBRRA-VTE (REGISTRATION: URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT05264168) and COBRRA-AF (NCT05256797) in 3 US health care insurance claims databases (Medicare, Optum Clinformatics, Merative MarketScan), the risk of bleeding with apixaban was evaluated relative to the risk with rivaroxaban. Pretreatment confounding factors were adjusted for via 1:1 propensity score matching. The primary outcome was a composite of major bleeding and clinically relevant non-major bleeding (defined using *International Classification of Diseases, Ninth Revision* and *Tenth Revision* diagnosis codes) recorded in the inpatient setting. Results across databases were pooled using fixed-effects meta-analyses. Protocols were preregistered on ClinicalTrials.gov before inferential analyses were conducted.

RESULTS: In the COBRRA-VTE emulation (N=14 023 matched pairs, mean age=67 years, 55% women), we observed a 31% decreased risk of bleeding for apixaban compared with rivaroxaban (incidence rate per 100 person-years: apixaban, 12.8; rivaroxaban, 19.1; hazard ratio, 0.69 [95% CI, 0.59–0.79]). For the COBRRA-AF emulation (N=176 990 matched pairs, mean age=74 years, 50% women), we similarly observed a 31% decreased risk for apixaban compared with rivaroxaban (incidence rate per 100 person-years: apixaban, 6.4; rivaroxaban, 9.2; hazard ratio, 0.69 [95% CI, 0.66–0.71]).

CONCLUSIONS: In database studies that closely emulated the design and measurements of the COBRRA-VTE and COBRRA-AF trials, a substantially lower risk of bleeding with apixaban was observed, consistent with the results of COBRRA-VTE (published after completion of our analyses), and indirect comparisons from other trials.

Key Words: comparative effectiveness research ■ factor xa inhibitors ■ observational study ■ pharmacoepidemiology ■ propensity score

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Venous thromboembolism (VTE) and atrial fibrillation (AF) are the 2 most common health conditions managed with oral anticoagulation therapies.¹ For

decades, the standard of care was vitamin K antagonists such as warfarin,² and the introduction of direct oral anticoagulants (DOACs) in 2010 transformed the

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WHAT IS KNOWN

- Published network meta-analyses and nonrandomized database studies, suggest that apixaban has a one-third lower risk of major and clinically relevant nonmajor bleeding compared with rivaroxaban in patients with atrial fibrillation and venous thromboembolism.
- Two randomized controlled trials comparing apixaban to rivaroxaban to assess the risk of major bleeding and clinically relevant nonmajor bleeding are currently ongoing.

WHAT THE STUDY ADDS

- Using the RCT-DUPLICATE (Randomized Controlled Trials Duplicated Using Prospective Longitudinal Insurance Claims: Applying Techniques of Epidemiology) design, data, and analytics framework, we conducted database studies to closely emulate the designs and measurements of the ongoing COBRRA-VTE (Comparison of Bleeding Risk Between Rivaroxaban and Apixaban for the Treatment of Acute Venous Thromboembolism) and COBRRA-AF trials (Comparison of Bleeding Risk Between Rivaroxaban and Apixaban in Patients With Atrial Fibrillation) to predict their results.
- Our results were concordant with previously published network meta-analyses and nonrandomized database studies.
- Our study highlights the role of well-designed database studies using fit-for-use data to generate timely results that complement randomized controlled trial evidence.

Nonstandard Abbreviations and Acronyms

AF	atrial fibrillation
DOAC	direct oral anticoagulant
HR	hazard ratio
PS	propensity score
RCT	randomized controlled trial
RCT-DUPLICATE	Randomized Controlled Trials Duplicated Using Prospective Longitudinal Insurance Claims: Applying Techniques of Epidemiology
VTE	venous thromboembolism

management of thromboembolic disorders. Apixaban and rivaroxaban target Factor Xa and have demonstrated improved safety and efficacy compared with vitamin K antagonists in multiple studies.^{3–7} These 2 drugs are the most frequently prescribed DOACs for VTE and AF.^{8–10}

In the absence of head-to-head randomized controlled trials (RCTs) comparing apixaban with rivaroxaban,

network meta-analyses of RCTs and several nonrandomized cohort studies conducted using health care databases consistently observed a clinically meaningful lower risk of major bleeding with apixaban versus rivaroxaban.^{11–16} Despite this body of evidence, investigators funded by the Ottawa Hospital Research Institute launched 2 RCTs to compare the safety of apixaban and rivaroxaban on the risk of major and clinically relevant nonmajor bleeding among patients with venous thromboembolism (COBRRA-VTE [Comparison of Bleeding Risk Between Rivaroxaban and Apixaban for the Treatment of Acute Venous Thromboembolism]) in 2017¹⁷ and AF (COBRRA-AF [Comparison of Bleeding Risk Between Rivaroxaban and Apixaban in Patients With Atrial Fibrillation]) in 2021.¹⁸ Both trials were ongoing at the time of this analysis (October 2024). The results from COBRRA-VTE were published in 2026,¹⁹ while COBRRA-AF is expected to be completed in December 2027.

The increasing availability of routinely collected and population-representative patient-level health care claims data provides opportunities to evaluate the comparative effectiveness and safety of drugs in nonrandomized studies. By applying modern study design and analysis approaches, researchers can generate evidence that complements and expands upon the findings from RCTs, enabling a better understanding of the effects of drugs in diverse clinical settings,²⁰ including frail older adults,²¹ patients with kidney disease,²² and those at elevated risk of bleeding.^{23,24} Moreover, robust evidence generated from well-designed nonrandomized database studies can contribute to evidence generation regarding clinical questions that may not be answered with RCTs. The RCT-DUPLICATE (Randomized Controlled Trials Duplicated Using Prospective Longitudinal Insurance Claims: Applying Techniques of Epidemiology) initiative systematically conducted emulations of 32 RCTs that were meant for regulatory submissions to evaluate when and how database studies can come to causal conclusions, using concordance with RCT results as a benchmark.^{25–30}

As part of the RCT-DUPLICATE initiative, we emulated the design and measurements of the COBRRA-VTE and COBRRA-AF trials using US health care claims databases. The goal was to predict the findings of the 2 COBRRA trials, which were ongoing and without publicly available results at the time the study protocols were preregistered.

METHODS

Data Sources

Data sources used for this study are available from the Centers for Medicare & Medicaid Services, Optum's de-identified Clinformatics Data Mart Database (Clinformatics), and Merative MarketScan. Access restrictions apply to these data, which

were used under license for this study. For both COBRRA-VTE and COBRRA-AF emulations, we used 3 US health care claims data sources: 2 commercial health insurance claims databases, Optum's de-identified Clinformatics Data Mart Database (Clinformatics; 2004–2020) and IBM MarketScan (2003–2019), as well as subsets of Medicare Parts A, B, and D (2009–2017), including all patients dispensed an oral anticoagulant. Each data source contained de-identified information for covered health care encounters of patients enrolled in participating health care insurance plans. The available data included demographic factors (age, sex), start and end dates of enrollment, dispensed medications with dates, dose, and days supply, and procedures and medical diagnoses with place of service and associated service dates. The date of death was captured with high completeness in Medicare data from the Master Beneficiary Summary File or the Vital Status File. In the 2 commercial databases, death was captured from the Social Security Death Master Index and in-hospital deaths. The study was approved by the Mass General Brigham institutional review board (2018P002966).

Study Design, Outcome, and Follow-Up

We identified groups of patients who were at least 18 years old and initiated apixaban or rivaroxaban starting from the apixaban approval date for the indication of VTE (August 21, 2014, for COBRRA-VTE) and AF (December 28, 2012, for COBRRA-AF) until the end of data availability in each database. The new-user active comparator cohort design used in this emulation has various strengths when analyzing secondary health care data, which have been described in detail elsewhere.^{31–33} New use was defined by a dispensing of apixaban or rivaroxaban after a washout of at least 180 days with medical and drug coverage and no dispensing of any DOAC or warfarin. The cohort entry date was the day of the first filled prescription of apixaban or rivaroxaban meeting both washout criteria and all other eligibility criteria. Patients were included if they had an inpatient or emergency department diagnosis of venous thromboembolism within 14 days before or on cohort entry (COBRRA-VTE) or diagnoses of AF in any care setting within 180 days before or on cohort entry (COBRRA-AF). These conditions were identified using the *International Classification of Diseases, Ninth Revision and Tenth Revision* codes (Table S1). Patients with any of the following characteristics in the 180 days before or on cohort entry were excluded: previous anticoagulation therapy, stage 4 or 5 chronic kidney disease, end-stage renal disease, dialysis, occurrence of pregnancy, cancer, or any other contraindication to anticoagulation (Figure 1). The eligibility criteria applied to the cohort were based on those of the COBRRA trials (Table S1).

The primary outcome was a composite of major bleeding and clinically relevant nonmajor bleeding using *International Classification of Diseases, Ninth Revision and Tenth Revision* diagnosis codes occurring in any position in the inpatient setting. All outcome definitions were based on previously validated algorithms with reported positive predictive values ranging from 86% to 97%.^{34,35} We modified the major bleeding algorithm in our study by adding clinically relevant nonmajor bleeding, defined by inpatient *International Classification of Diseases* diagnosis codes for noncritical sites without transfusion (Table S2). In line with the RCTs, the COBRRA-VTE and COBRRA-AF emulations assessed the outcome during

the first 3-month and 12-month follow-up periods after initiation of anticoagulation therapy, respectively. The primary analysis used an as-treated approach regarding follow-up wherein study follow-up began the day after cohort entry and continued until the end of the study period, the end of continuous health plan enrollment, the occurrence of a study outcome, discontinuation of the index medication (with a 10-day allowable grace period and risk window), a switch to the comparator, initiation of another anticoagulant, admission to a nursing home, or death. The duration of the index exposure was captured using the date of dispensation and the days supply. A 10-day gap was allowed between the end of a prior dispensation and a subsequent one to allow for nonadherence and the potential continued biologic effect of exposure after the last pill was taken. Similarly, the duration of exposure was extended 10 days beyond the last dispensation in a sequence. This approach emulates a per-protocol analysis in the RCTs and minimizes exposure misclassification. For the secondary as-started analysis, emulating an intention-to-treat analysis of the RCTs, we did not censor for medication discontinuation, switching, or addition of other anticoagulants.

Statistical Analysis

Within each database, we matched apixaban initiators with rivaroxaban initiators using 1:1 propensity score (PS) matching without replacement on their estimated PS using a nearest-neighbor approach with a caliper of 1% to balance >100 preexposure patient characteristics. PSs were estimated via a multivariable logistic regression model incorporating all measured baseline covariates to predict the probability of receiving apixaban versus rivaroxaban. The covariates were measured during the 6 months before and including the cohort entry date and included demographics, calendar time, comorbidities, use of medications, indicators of health care utilization, as well as indexes of overall health state such as the combined comorbidity score and claims-based frailty index.^{36,37} All covariates included in the PS are shown in Tables S3 and S4. Data analyses were conducted separately in each database, and results were pooled using fixed-effects meta-analyses.

After PS matching, we estimated incidence rates of the primary outcomes for each treatment group, including their 95% CIs, and fit Cox proportional hazards models to estimate hazard ratios (HRs) with their 95% CIs. In addition, we plotted the cumulative incidence of major and clinically relevant nonmajor bleeding (Figure 2). Trial emulations were conducted using the validated Aetion Evidence Platform^{38,39} with supplemental programming using SAS 9.4 and CRAN R, v4.1.1.

Development of the Study Protocol

Several feasibility and data checks were conducted during the development of the study protocol, before preregistration on ClinicalTrials.gov.⁴⁰ These checks included assessing whether there was at least equal statistical power in the database study as in the relevant COBRRA trial for detecting superiority of the primary end point. The statistical power was estimated from the total number of event counts without stratifying by treatment group. In addition, we assessed the pre- and postmatching balance of baseline covariates between the study arms by calculating standardized differences, reviewing the overlap in the PS distributions of the 2 arms, and the C statistics before

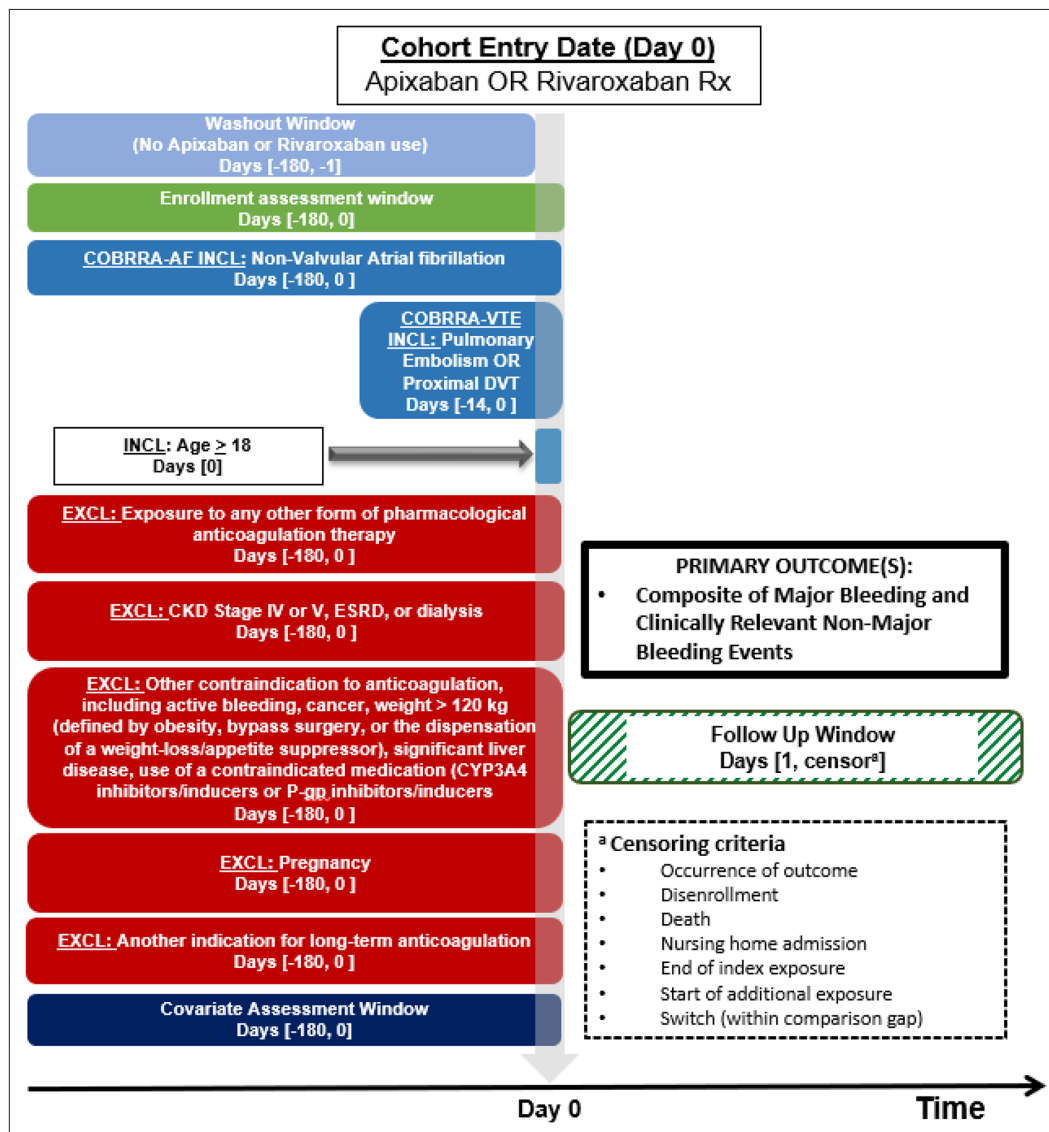


Figure 1. Schematic of the study design.

CKD indicates chronic kidney disease; COBRRA-AF, Comparison of Bleeding Risk Between Rivaroxaban and Apixaban in Patients With Atrial Fibrillation; COBRRA-VTE, Comparison of Bleeding Risk Between Rivaroxaban and Apixaban for the Treatment of Acute Venous Thromboembolism; DVT, deep vein thrombosis; ESRD, end stage renal disease; EXCL, exclusion; INLC, inclusion; and Rx, prescription.

and after PS matching (see Appendix for Tables S3 through S6). All details of these data and feasibility checks are included in the study protocols registered at ClinicalTrials.gov before the inferential analyses were started (NCT05264168 registered on March 3, 2022; NCT05256797 registered on February 25, 2022).

To contextualize our findings, we compared results for the primary outcome of major and clinically relevant nonmajor bleeding in our study with existing network meta-analyses of RCTs and the results of the COBRRA-VTE trial, which were published after completion of the analyses. Agreement between the trial and database study estimates was evaluated using 3 prespecified binary metrics previously used in the RCT-DUPLICATE initiative²⁵:

1. Statistical agreement, defined as both the database study and trial estimates and their 95% CIs lying on the same side of the null;

2. Estimate agreement, defined as the estimate of the database study falling within the 95% CI of the trial;
3. Standardized difference agreement, defined as an absolute standardized difference $|Z| < 1.96$, with $z = \frac{\hat{\theta}_{RCT} - \hat{\theta}_{RWE}}{\sqrt{\hat{\sigma}_{RCT}^2 + \hat{\sigma}_{RWE}^2}}$ where $\hat{\theta}_{RCT}$ and $\hat{\theta}_{RWE}$ are the treatment effect estimates with associated variances $\hat{\sigma}^2$.

RESULTS

COBRRA-VTE Emulation

In the COBRRA-VTE emulation, we identified 41 337 eligible patients with VTE, initiating either apixaban (N=17 913) or rivaroxaban (N=23 424). In the unmatched cohort, patients initiating apixaban were older than rivaroxaban initiators (mean age [SD]: 68.0 [14.3]

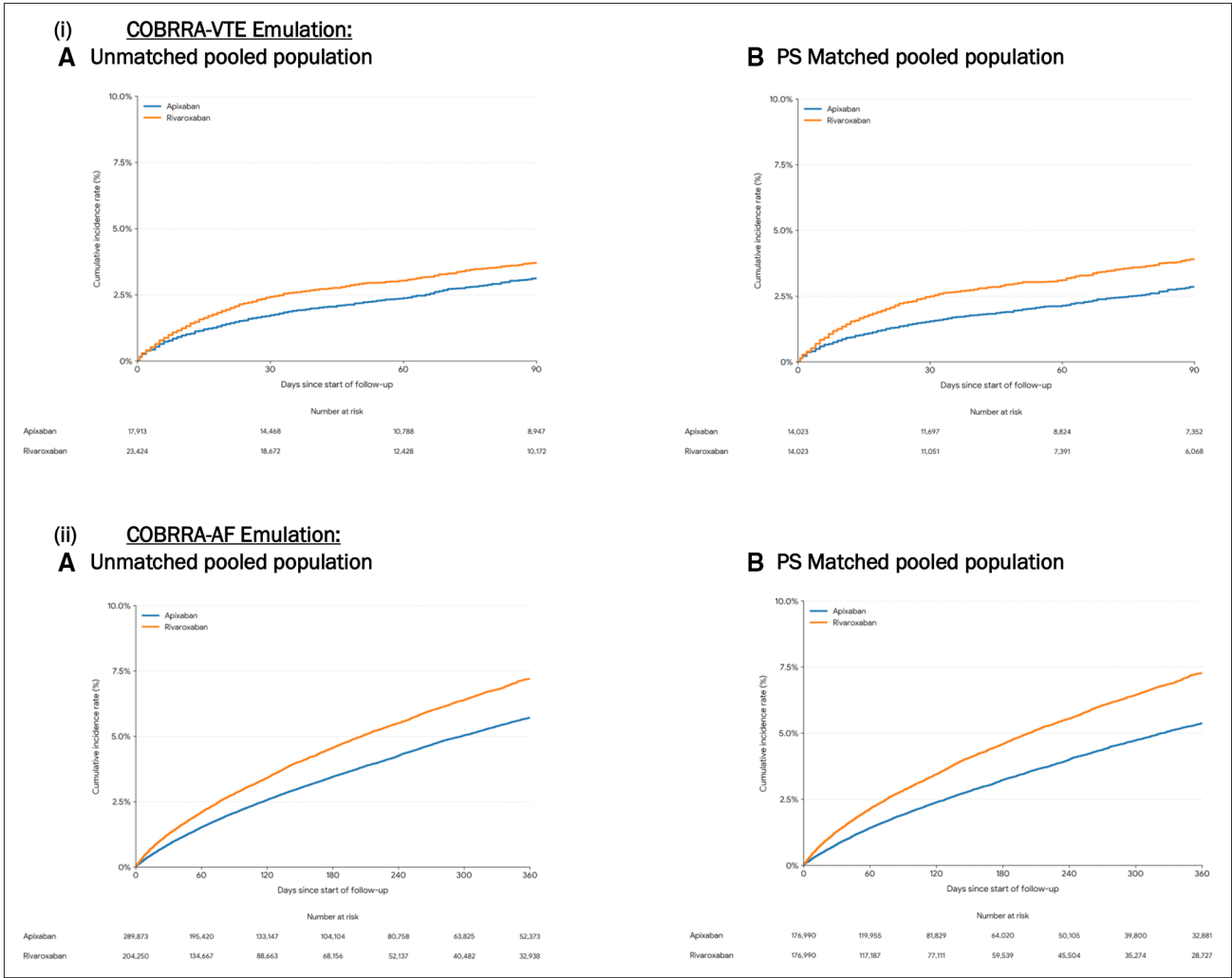


Figure 2. Cumulative incidence of the composite major bleeding and clinically relevant nonmajor bleeding outcome before and after propensity score matching for both COBRRA-VTE (Comparison of Bleeding Risk Between Rivaroxaban and Apixaban for the Treatment of Acute Venous Thromboembolism) and COBRRA-AF (Comparison of Bleeding Risk Between Rivaroxaban and Apixaban in Patients With Atrial Fibrillation) emulations.

A new-user active comparator design with propensity score (PS) matching was used to estimate treatment effects across 2 emulations: COBRRA-VTE (i): In the PS-matched population, curves diverged early, with apixaban showing a lower incidence over a 90-day follow-up period (N=14 023 per group). COBRRA-AF (ii): In the PS-matched population, the lower cumulative incidence for apixaban was maintained over a 365-day follow-up period (N=176 990 per group). The results consistently indicate that apixaban was associated with a lower cumulative incidence of outcome events relative to rivaroxaban in both unmatched (A) and PS-matched pooled (B) populations.

years versus 64.5 [14.6] years), and they had a slightly higher mean combined comorbidity score (mean score [SD]: 2.82 [2.28] versus 2.09 [2.04]). The PS-matched population had 14 023 matched pairs with well-balanced baseline characteristics (Table 1 and Table S3) and similar PS distributions, with C statistics in all 3 databases moving from 0.7 to 0.5 after matching (Table S5). Baseline characteristics of the matched cohorts for each database are included in Table S7. The median (interquartile range) follow-up time was 79 (38–90) days. Details of follow-up time and censoring reasons in each database are presented in Table S8.

In the primary as-treated analysis, the incidence rate for the composite major bleeding and clinically relevant

nonmajor bleeding outcome was 12.8 per 100 person-years (320 events) among apixaban initiators and 19.1 per 100 person-years (440 events) among rivaroxaban initiators. We observed a 31% decreased hazard (HR, 0.69 [95% CI, 0.59–0.79]) or 6.3 fewer events (RD –6.3 [95% CI, –8.6 to –4.0]) per 100 person-years with apixaban use versus rivaroxaban (Table 2 and Figure 3). In the individual databases, the results were as follows: Optum Clinformatics, HR=0.71 (95% CI, 0.52–0.97); MarketScan, HR=0.90 (95% CI, 0.62–1.31); and Medicare, HR=0.65 (95% CI, 0.54–0.78; Table S9). In the intention-to-treat analysis that followed patients for up to a year, the matched population had 14 079 patients in the apixaban and rivaroxaban

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Table 1. COBRRA-VTE and COBRRA-AF Trial Emulation–Selected* Baseline Characteristics of Patients After 1:1 Propensity Score Matching

Patient characteristics	COBRRA-VTE emulation			COBRRA-AF emulation		
	Apixaban, N=14 023	Rivaroxaban, N=14 023	St. diff	Apixaban, N=176 990	Rivaroxaban, N=176 990	St. diff
Age, y; mean (SD)	66.77 (14.25)	66.77 (13.85)	0.00	73.88 (9.75)	73.91 (9.65)	0.00
Age categories, n (%)						
18–54	2964 (21.1%)	2927 (20.9%)	0.00	7970 (4.5%)	7885 (4.5%)	0.00
55–64	2244 (16.0%)	2276 (16.2%)	0.01	18 361 (10.4%)	18 391 (10.4%)	0.00
65–74	3864 (27.6%)	3884 (27.7%)	0.00	62 360 (35.2%)	62 366 (35.2%)	0.00
≥75	4951 (35.3%)	4936 (35.2%)	0.00	88 299 (49.9%)	88 348 (49.9%)	0.00
Males sex; n (%)	6315 (45.0%)	6273 (44.7%)	0.01	88 972 (50.3%)	88 932 (50.2%)	0.00
Race; n (%)						
White	8209 (58.5%)	8254 (58.9%)	0.01	127 949 (72.3%)	127 937 (72.3%)	0.00
Black	1377 (9.8%)	1334 (9.5%)	0.01	7011 (4.0%)	7016 (4.0%)	0.00
Hispanic	355 (2.5%)	340 (2.4%)	0.01	4162 (2.4%)	4140 (2.3%)	0.01
Region; n (%)						
Northeast	2488 (17.7%)	2516 (17.9%)	0.01	33 033 (18.7%)	32 987 (18.6%)	0.00
South	6178 (44.1%)	6135 (43.7%)	0.01	69 779 (39.4%)	69 835 (39.5%)	0.00
Midwest	3123 (22.3%)	3188 (22.7%)	0.01	40 027 (22.6%)	40 214 (22.7%)	0.00
West	2213 (15.8%)	2165 (15.4%)	0.01	33 957 (19.2%)	33 746 (19.1%)	0.00
Burden of comorbidities in past 180 d						
Combined comorbidity score†; mean (SD)	2.48 (2.11)	2.48 (2.12)	0.00	1.89 (1.86)	1.90 (1.86)	0.01
Frailty score‡; mean (SD)	0.19 (0.07)	0.19 (0.07)	0.00	0.18 (0.06)	0.18 (0.06)	0.00
CHADS2 score; mean (SD)	0.76 (0.97)	0.75 (0.95)	0.01	1.22 (1.18)	1.22 (1.18)	0.00
Cardiovascular comorbidities observed in past 180 d; n (%)						
Diabetes without mention of complications	2816 (20.1%)	2820 (20.1%)	0.00	40 852 (23.1%)	40 863 (23.1%)	0.00
Hypertension	9575 (68.3%)	9565 (68.2%)	0.00	141 665 (80.0%)	141 738 (80.1%)	0.00
Hyperlipidemia	6677 (47.6%)	6648 (47.4%)	0.00	91 604 (51.8%)	91 776 (51.9%)	0.00
Acute myocardial infarction	663 (4.7%)	653 (4.7%)	0.00	3895 (2.2%)	3909 (2.2%)	0.00
ACS/unstable angina	460 (3.3%)	469 (3.3%)	0.00	3729 (2.1%)	3737 (2.1%)	0.00
Old Myocardial Infarction	575 (4.1%)	569 (4.1%)	0.00	6393 (3.6%)	6490 (3.7%)	0.01
Stable angina	327 (2.3%)	347 (2.5%)	0.01	5820 (3.3%)	5860 (3.3%)	0.00
Coronary atherosclerosis and other forms of chronic ischemic heart disease	2343 (16.7%)	2319 (16.5%)	0.01	42 366 (23.9%)	42 366 (23.9%)	0.00
Previous cardiac procedure [Coronary artery bypass grafting or Percutaneous transluminal coronary angioplasty or stent] + history of CABG or PTCA	766 (5.5%)	755 (5.4%)	0.00	10 845 (6.1%)	10 930 (6.2%)	0.00
Ischemic stroke	892 (6.4%)	879 (6.3%)	0.00	18 172 (10.3%)	18 262 (10.3%)	0.00
Hemorrhagic stroke	28 (0.2%)	25 (0.2%)	0.00	258 (0.1%)	275 (0.2%)	0.03
Transient ischemic attack	243 (1.7%)	244 (1.7%)	0.00	6538 (3.7%)	6588 (3.7%)	0.00
Other cerebrovascular disease	417 (3.0%)	397 (2.8%)	0.01	5483 (3.1%)	5492 (3.1%)	0.00
Late effects of cerebrovascular disease	334 (2.4%)	325 (2.3%)	0.01	4227 (2.4%)	4294 (2.4%)	0.00
Congestive heart failure	1995 (14.2%)	1947 (13.9%)	0.01	32 588 (18.4%)	32 693 (18.5%)	0.00
Peripheral vascular disease (PVD) or PVD surgery	1095 (7.8%)	1099 (7.8%)	0.00	11 198 (6.3%)	11 285 (6.4%)	0.00
Other cardiac dysrhythmia	2069 (14.8%)	2065 (14.7%)	0.00	132 861 (75.1%)	132 733 (75.0%)	0.00
Cardiac conduction disorders	1119 (8.0%)	1116 (8.0%)	0.00	13 819 (7.8%)	13 707 (7.7%)	0.00
Other cardiovascular disease events	3814 (27.2%)	3840 (27.4%)	0.00	53 825 (30.4%)	53 919 (30.5%)	0.00
Cardiovascular medications used in past 180 d; n (%)						
Angiotensin-converting enzyme inhibitors	3359 (24.0%)	3345 (23.9%)	0.00	54 498 (30.8%)	54 500 (30.8%)	0.00
Angiotensin receptor blockers	2203 (15.7%)	2269 (16.2%)	0.01	41 663 (23.5%)	41 567 (23.5%)	0.00

(Continued)

Table 1. Continued

Patient characteristics	COBRRA-VTE emulation			COBRRA-AF emulation		
	Apixaban, N=14 023	Rivaroxaban, N=14 023	St. diff	Apixaban, N=176 990	Rivaroxaban, N=176 990	St. diff
α-Blockers	1227 (8.7%)	1186 (8.5%)	0.01	20 488 (11.6%)	20 444 (11.6%)	0.00
β-Blockers	3907 (27.9%)	3920 (28.0%)	0.00	119 643 (67.6%)	119 713 (67.6%)	0.00
Loop diuretics	1877 (13.4%)	1876 (13.4%)	0.00	37 654 (21.3%)	37 713 (21.3%)	0.00
Other diuretics§	367 (2.6%)	384 (2.7%)	0.01	7906 (4.5%)	7895 (4.5%)	0.00
Use of nitrates	512 (3.7%)	513 (3.7%)	0.00	9650 (5.5%)	9594 (5.4%)	0.00
Use of statins	4933 (35.2%)	4928 (35.1%)	0.00	90 874 (51.3%)	90 893 (51.4%)	0.00
Use of other lipid-lowering drugs	598 (4.3%)	605 (4.3%)	0.00	7385 (4.2%)	7287 (4.1%)	0.01
Calcium channel blockers	3023 (21.6%)	3064 (21.8%)	0.00	67 972 (38.4%)	68 026 (38.4%)	0.00
Digoxin	66 (0.5%)	63 (0.4%)	0.01	13 781 (7.8%)	13 826 (7.8%)	0.00
Antiarrhythmic	55 (0.4%)	65 (0.5%)	0.01	31 224 (17.6%)	31 299 (17.7%)	0.00
Measures of health care utilization during past 180 d						
No. of hospitalizations; mean (SD)	1.23 (0.60)	1.23 (0.60)	0.00	0.33 (0.58)	0.33 (0.59)	0.00
No. of emergency department (ED) visits; mean (SD)	2.34 (3.99)	2.31 (5.08)	0.01	0.87 (1.70)	0.88 (2.90)	0.00
No. of office visits; mean (SD)	8.02 (9.06)	8.00 (13.09)	0.00	9.75 (9.43)	9.71 (15.73)	0.00
No. of distinct medications; mean (SD)	9.00 (5.27)	9.01 (7.85)	0.00	8.62 (4.38)	8.63 (7.28)	0.00
No. of cardiologist visits; mean (SD)	2.13 (3.67)	2.12 (5.28)	0.00	4.50 (5.23)	4.49 (9.39)	0.00
No. of internal medicine/family medicine visits; mean (SD)	9.51 (10.98)	9.53 (16.90)	0.00	7.74 (10.20)	7.74 (16.60)	0.00
Bone mineral density; n (%)	442 (3.2%)	450 (3.2%)	0.00	6290 (3.6%)	6356 (3.6%)	0.00
Colonoscopy; n (%)	511 (3.6%)	523 (3.7%)	0.01	7253 (4.1%)	7263 (4.1%)	0.00
Flu vaccine; n (%)	2950 (21.0%)	2981 (21.3%)	0.01	46 584 (26.3%)	46 627 (26.3%)	0.00
Mammogram; n (%)	1211 (8.6%)	1196 (8.5%)	0.00	16 443 (9.3%)	16 518 (9.3%)	0.00
Prostate Specific Antigen test or Prostate exam for Digital Rectal Exam ; n (%)	1237 (8.8%)	1244 (8.9%)	0.00	22 517 (12.7%)	22 417 (12.7%)	0.00
Pap smear; n (%)	497 (3.5%)	501 (3.6%)	0.01	3860 (2.2%)	3877 (2.2%)	0.00
Copay per prescription¶; mean (SD), \$	16.90 (29.02)	17.07 (46.83)	0.00	79.94 (99.03)	79.79 (180.57)	0.00
Copay for pharmacy cost#; mean (SD), \$	169.97 (337.36)	171.10 (478.78)	0.00	183.69 (303.14)	183.19 (507.28)	0.00
Insurance type; n (%)						
Commercial	3944 (28.1%)	3961 (28.2%)	0.00	23 524 (13.3%)	23 518 (13.3%)	0.00
Medicare	2695 (19.2%)	2678 (19.1%)	0.00	41 113 (23.2%)	41 119 (23.2%)	0.00
Low-income indicator**; n (%)	472 (3.4%)	451 (3.2%)	0.01	4108 (2.3%)	4142 (2.3%)	0.00

*Full tables of baseline characteristics provided in [Tables S3](#) and [S4](#).

†Gagne et al.³⁶

‡Kim et al.³⁷

§Other diuretics include: eplerenone, spironolactone and triamterene.

¶Average copay amount per prescription for each patient.

#Total copay amounts for all pharmacy-related services (based on all prescription claims) for each patient.

**This indicates beneficiaries who receive the Low-income subsidy, including those enrolled in Medicare Advantage Prescription Drug (MAPD) plan as well those enrolled in an MAPD plan and are eligible for both Medicare and Medicaid (Dual eligible).

arms, and we observed a 29% decreased risk of major or clinically relevant nonmajor bleeding (HR, 0.71 [95% CI, 0.62–0.81]; [Table S10](#)).

COBRRA-AF Emulation

In the COBRRA-AF emulation, after applying the eligibility criteria, we identified 495 701 eligible patients with AF initiating either apixaban (N=291 020) or rivaroxaban (N=204 681). In our cohort, compared with rivaroxaban initiators, patients initiating apixaban were slightly older (mean age [SD]: 75.2 [9.8] years versus

73.2 [9.9] years) and had a higher mean combined comorbidity score (mean score [SD]: 2.14 [2.04] versus 1.84 [1.83]). In the PS-matched population, there were 33 730, 30 907, and 112 353 matched pairs within the Optum Clinformatics, MarketScan, and Medicare databases, respectively. Baseline characteristics of the matched cohorts for each database are included in [Table S11](#). In the pooled cohort (N=176 990 apixaban and rivaroxaban matched pairs) baseline characteristics were well balanced ([Table 1](#) and [Table S4](#)), with similar PS distributions. The C statistics in the Optum Clinformatics and MarketScan databases moved from

Table 2. Event Rates and Effect Estimates for As-Treated Analyses of Primary Outcome* and 2 Secondary Outcomes†,‡

Trial name	Apixaban			Rivaroxaban			Rate difference/100 person-years (95% CI)	Adjusted HR (95% CI)
	N	Events	Incidence rate/100 person-years (95% CI)	N	Events	Incidence rate/100 person-years (95% CI)		
Primary outcome: major and clinically relevant nonmajor bleeding (in any position) after 1:1 PS-matching								
COBRRA-VTE	14 023	320	12.8 (11.5–14.3)	14 023	440	19.1 (17.4–21.0)	−6.3 (−8.6 to −4.0)	0.69 (0.59–0.79)
COBRRA-AF	176 990	4865	6.4 (6.2–6.6)	176 990	6647	9.2 (9.0–9.5)	−2.8 (−3.1 to −2.5)	0.69 (0.66–0.71)
Secondary outcome I: major bleeding (in any position) after 1:1 PS-matching								
COBRRA-VTE	14 023	118	4.7 (3.9–5.6)	14 023	184	7.9 (6.8–9.2)	−3.2 (−4.7 to −1.8)	0.59 (0.47–0.75)
COBRRA-AF	176 990	1860	2.4 (2.3–2.5)	176 990	2891	4.0 (3.8–4.1)	−1.6 (−1.7 to −1.4)	0.60 (0.56–0.64)
Secondary outcome II: (i) recurrent VTE for COBRRA-VTE and (ii) stroke for COBRRA-AF after 1:1 PS-matching								
COBRRA-VTE	14 023	663	27.0 (25.0–29.1)	14 023	694	30.5 (28.3–32.9)	−3.5 (−6.6 to −0.5)	0.91 (0.82–1.02)
COBRRA-AF	176 990	695	0.9 (0.8–1.0)	176 990	782	1.1 (1.0–1.2)	−0.2 (−0.3 to −0.1)	0.94 (0.84–1.04)

COBRRA-AF indicates Comparison of Bleeding Risk Between Rivaroxaban and Apixaban in Patients With Atrial Fibrillation; COBRRA-VTE, Comparison of Bleeding Risk Between Rivaroxaban and Apixaban for the Treatment of Acute Venous Thromboembolism; HR, hazard ratio; and PS, propensity score.

*Major and clinically relevant nonmajor bleeding (in any position).

†Major bleeding (in any position).

‡Recurrent VTE for COBRRA-VTE and stroke for COBRRA-AF after 1:1 PS-matching.

0.7 to 0.5 after matching; they moved from 0.6 to 0.5 after matching in the Centers for Medicare & Medicaid Services database (Table S6). The median (interquartile range) for follow-up time was 98 (38–222) days. Details of follow-up time and censoring reasons in each database are presented in Table S12.

In the primary as-treated analysis, the incidence rate for the composite major bleeding and clinically relevant nonmajor bleeding outcome was 6.4 per 100 person-years (4865 events) among apixaban initiators and 9.2 per 100 person-years (6647 events) among rivaroxaban initiators. Thus, we observed a 31% decreased hazard (HR, 0.69 [95% CI, 0.66–0.71]) or 2.8 fewer events (RD, −2.8 [95% CI, −3.1 to −2.5]) per 100 person-years with exposure to apixaban versus rivaroxaban (Table 2 and Figure 3). Risk reductions for major and clinically relevant nonmajor bleeding were consistent across the

individual databases: Optum Clinformatics, HR=0.77 (95% CI, 0.70–0.84); MarketScan, HR=0.71 (95% CI, 0.64–0.80), and Medicare, HR=0.68 (95% CI, 0.66–0.72; Table S9). In the intention-to-treat analysis that followed patients for up to a year, there were 177 423 matched pairs, and we observed a 26% risk reduction (HR, 0.74 [95% CI, 0.72–0.77]; Table S13).

For the primary outcome of major and clinically relevant nonmajor bleeding in our study, our results were consistent with existing network meta-analyses of RCTs, showing a substantially lower risk of bleeding events for patients treated with apixaban (Figure 3). When compared with the results of the COBRRA-VTE trial (RCT, 0.46 [95% CI, 0.33–0.65]; database: 0.69 [95% CI, 0.59–0.79]), the findings were consistent in showing a strong protective effect for apixaban, and our results met the statistical significance agreement metric. However,

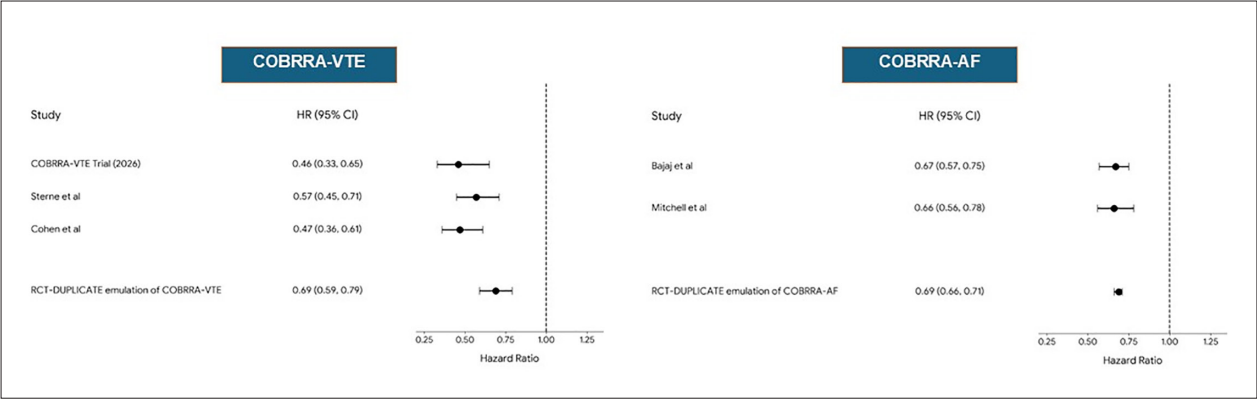


Figure 3. Effect estimates of COBRRA trial emulations vs network meta-analyses and COBRRA-VTE (Comparison of Bleeding Risk Between Rivaroxaban and Apixaban for the Treatment of Acute Venous Thromboembolism) comparing apixaban to rivaroxaban for the primary composite bleeding outcome.

Analyses indicate that the risk of bleeding was lower among apixaban initiators compared with rivaroxaban initiators in both COBRRA-VTE and COBRRA-AF Trial (Comparison of Bleeding Risk Between Rivaroxaban and Apixaban in Patients With Atrial Fibrillation) emulations. These findings were seen to be consistent when compared against network meta-analyses and the COBRRA-VTE trial results. HR indicates hazard ratio.

because the observed effect was stronger in the trial than in clinical practice, and the point estimate of the database study was slightly outside the narrow CIs of the trial, neither the estimate agreement metric nor the standardized difference agreement metric was met.

DISCUSSION

We emulated the designs of 2 ongoing randomized trials that directly compare apixaban with rivaroxaban regarding the risk of major bleeding and clinically relevant nonmajor bleeding in patients with venous thromboembolism (COBRRA-VTE) and AF (COBRRA-AF) to predict the results of these trials. Our studies were conducted using nonrandomized health care claims database analyses. Before preregistering the study protocols, we demonstrated evidence of adequate statistical power and the ability to achieve balance on all relevant confounders. Such data checks were conducted to understand the data quality and completeness and to assess our ability to control confounding before engaging in the actual prediction of the 2 ongoing COBRRA trials. We observed a one-third lower risk of major and clinically relevant nonmajor bleeding in apixaban initiators compared with rivaroxaban initiators with VTE and AF.

Although database studies are recognized as an approach to generate evidence regarding the safety and efficacy of medications in routine clinical practice, some stakeholders continue to be skeptical about whether these types of studies can generate valid causal conclusions. The view that studies that are not randomized are intractably biased^{41–43} is a consequence of the prevalence of demonstrably flawed database studies, resulting in difficulty judging the quality of a given database study.^{26,44,45} Such circumstances lead to situations such as the randomized head-to-head DOAC comparisons in the COBRRA trials where there is a plethora of evidence from network meta-analyses and observational studies with low risk of bias that consistently demonstrate the reduction in risk of bleeding for apixaban versus rivaroxaban.

Our database study predicting the results of the COBRRA-VTE trial provided evidence of substantially lower risk of major bleeding with apixaban, similar to the trial. The results of COBRRA-AF are not yet available for comparison. However, our database study results are also consistent with existing network meta-analyses of RCTs that also observed strong protective effects of apixaban versus rivaroxaban. For the primary outcome of major and clinically relevant nonmajor bleeding, 2 network meta-analyses of RCTs comparing apixaban with rivaroxaban in patients with AF^{11,12} and 2 network meta-analyses in patients with VTE^{13,14} obtained effect estimates of similar magnitude to our COBRRA emulations.

In spite of this body of evidence, as noted by COBRRA trial investigators on ClinicalTrials.gov, there have been no direct head-to-head comparisons of these 2

anticoagulants, meaning comparative safety data are not available. “Increasing use of DOACs ... highlight the need for a comparison trial ...”⁴⁶ (emphasis added). The risk of major and clinically relevant nonmajor bleeding from rivaroxaban in the ROCKET-AF⁴ (Rivaroxaban Once Daily Oral Direct Factor Xa Inhibition Compared with Vitamin K Antagonism for Prevention of Stroke and Embolism Trial in Atrial Fibrillation) trial was 14.9% per year, and assuming that the relative risk of bleeding for apixaban versus rivaroxaban is ≈ 0.66 , as has been consistently shown by multiple network meta-analyses and nonrandomized database studies, this would result in a risk of approximately a 9.8% per year for apixaban. This translates to an estimated number needed to harm of 20 and a projected 75 excess bleeding events in the COBRRA-AF trial.

Regulatory authorities are increasingly considering nonrandomized studies as part of a continuum of types of study designs that can be used together in evaluating drug safety and effectiveness.^{47–49} Such approaches can encourage the appropriate use of nonrandomized study designs to complement evidence from RCTs, accelerate evidence development, and help reduce research waste.

There are limitations to the specific trial emulations reported in this study. First, there are differences in the measurement and ascertainment of study parameters between those collected during the trial as compared with definitions applied in the database study. In the absence of a publicly available protocol or other information about how major and clinically relevant nonmajor bleeding were defined for the COBRRA trials, our team had to make a decision and chose to base our outcome algorithm on 1 previously used by Food and Drug Administration (FDA) investigators that was based on inpatient diagnoses in critical sites or inpatient diagnoses in noncritical sites with procedure codes for transfusion (Table S2).^{34,35} Although this algorithm was found to have a high positive predictive value, it has various limitations in capturing major bleeding, such as requiring a primary hospital discharge diagnosis and thus not detecting some bleeding events coded as a secondary diagnosis. Also, the specific sites and bleeding definition may or may not align with the definitions used by the COBRRA investigative team. Prior work has demonstrated the sensitivity of results for studies of direct oral anticoagulants to the definition of major bleeding.⁴⁸ Second, as with all observational research, residual confounding due to some unmeasured characteristics cannot be completely ruled out. However, our prespecified data checks demonstrated that the new-user active comparator study design equipped with PS-adjustment was able to balance >100 clinically relevant variables, substantially reducing concerns of confounding by unmeasured covariates.

CONCLUSIONS

In this large cohort study designed to predict the findings of 2 ongoing DOAC trials (COBRRA-VTE and

COBRRA-AF) before their completion, we observed that apixaban was meaningfully safer than rivaroxaban with regard to a composite bleeding outcome in patients with AF and VTE. The observed decreased risk of bleeding with apixaban compared with rivaroxaban is consistent with the results from COBRRA-VTE and indirect comparisons from other RCTs. When the results of the COBRRA-AF trial become available, we will assess the degree to which our emulation was successful in predicting the trial findings and attempt to understand any observed divergence in results.

ARTICLE INFORMATION

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Author Contributions

Drs Mahesri, Schneeweiss, Lin, L. Zabolka, Drs Concato, Bradley and Wang contributed to the conception and design of the study. Dr Mahesri, L. Zabolka, and Dr Wang analyzed the data. Drs Mahesri and Wang wrote the first draft of the article. All authors critically revised the article and approved the final article.

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Disclosures

Dr Wang has been an ad hoc consultant to Veracity Healthcare Analytics, Exponent, Inc, and MITRE an FFRDC for the Centers for Medicare and Medicaid Services for unrelated work. Sebastian Schneeweiss is principal investigator of the US Food and Drug Administration (FDA) Sentinel Innovation Center funded by the FDA, coprincipal investigator of an investigator-initiated grant to the Brigham and Women's Hospital from Boehringer Ingelheim and UCB Pharma unrelated to the topic of this study. He is a consultant to Aetion, Inc, a software manufacturer of which he owns equity. His interests were declared, reviewed, and approved by the Brigham and Women's Hospital and MGB HealthCare System in accordance with their institutional compliance policies. The other authors report no conflicts.

Transparency Statement

This study was preregistered before conducting analyses at ClinicalTrials.gov (NCT05264168 and NCT05256797). The initial protocol and subsequent amendments are also included in the ClinicalTrials.gov submission. All analyses for these trial emulations were conducted using the validated Aetion Evidence Platform and the complete reports are available at <https://osf.io/dvne9/>. The study used de-identified data covered by data use agreements and received approval from the Mass General Brigham institutional review board (2018P002966). Access to the de-identified source data may be granted to authorized researchers via application and licensing with the data providers: Medicare (resdac@umn.edu, <https://resdac.org/research-identifiable-files-rif-requests>), Optum's de-identified Clinformatics Data Mart Database (Clinformatics; connected@optum.com, <https://www.optum.com/business/solutions/life-sciences/real-world-data/claims-data.html>) and Merative MarketScan (<https://www.merative.com/contact/>). The article is aligned with guidance from the Strengthening the Reporting of Observational Studies in Epidemiology checklist.

Supplemental Material

Tables S1–S13

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